

Erdős Problem #857

EP-857

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $m = m(n, k)$ be minimal such that in any collection of sets $A_1, \dots, A_m \subseteq \{1, \dots, n\}$ there must exist a sunflower of size k - that is, some collection of k of the A_i which pairwise have the same intersection. Estimate $m(n, k)$, or even better, give an asymptotic formula.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>